

Dual-Mode Hidden Surveillance Detection System Using ESP32 and AD8318

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Abstract—This paper presents a Dual-Mode Hidden Surveillance Detection System that combines infrared (IR) and radiofrequency (RF) sensing to detect hidden surveillance devices in indoor environments. An ESP32 microcontroller serves as the central processing unit, interfacing with an AD8318 RF logarithmic power detector for RF monitoring in the 1 GHz to 2.4 GHz band and a TSOP1838 infrared receiver module for IR-based detection. The RF subsystem targets common wireless surveillance technologies such as GSM links, Wi-Fi cameras, Bluetooth-based cameras, and other low-power RF transmitters. The AD8318 converts RF power at its input to a proportional logarithmic output voltage, which is sampled by the ESP32's 12-bit analog-to-digital converter (ADC) and mapped to an approximate input power level in dBm, visualized as a real-time RF intensity bar on an OLED display. When the RF power level exceeds a calibrated threshold, the system displays “RF Camera Detected” on a 16×2 LCD and activates an audible buzzer. In parallel, the IR subsystem employs a demodulating IR receiver tuned around 38 kHz to sense infrared emissions from night-vision-equipped cameras. The dual-mode approach provides complementary coverage, significantly reducing false positives compared to single-modality detectors. Experimental validation confirms detection ranges of 1–3 metres for typical surveillance devices, with alert response times below 500 ms and RF power accuracy within ± 2 dBm post-calibration, demonstrating the system's suitability for privacy protection in hotels, offices, and examination centres.

Index Terms—ESP32 microcontroller, RF detection, infrared detection, AD8318 logarithmic detector, hidden camera detection, surveillance countermeasures, embedded systems, dual-mode sensing, IoT security, OLED display

I. INTRODUCTION

The proliferation of low-cost miniature cameras, wireless transmission modules, and compact storage devices has signif-

icantly increased the risk of covert surveillance in domestic, commercial, and institutional spaces. Hidden cameras can be concealed in everyday objects such as smoke detectors, chargers, clocks, and lamps, and frequently leverage wireless communication technologies including Wi-Fi, Bluetooth, or cellular links to stream or upload captured video content [6].

Traditional physical inspection techniques are inadequate because modern surveillance devices often employ sophisticated camouflage and may be deployed in locations that are difficult to visually inspect. Electronic counter-surveillance tools have emerged that exploit the physical signatures of hidden devices, predominantly in the radio-frequency (RF) and infrared (IR) domains. Wireless cameras emit RF energy while transmitting, which can be detected using RF receivers or logarithmic detectors. Many night-vision cameras integrate IR LED arrays whose emissions can be detected using IR receivers [7].

A practical hidden surveillance detector must support reliable sensing of one or more of these modalities while operating with low power, compact electronics, and a user-friendly interface suitable for non-expert users. The ESP32 microcontroller provides a cost-effective solution by combining digital processing, analog acquisition, and communication capabilities in a single integrated device [3]. The AD8318 RF logarithmic detector enables accurate measurement of RF power across a wide dynamic range, making it well suited for RF-based surveillance detection [1].

Privacy violations from unauthorized surveillance threaten personal security, intellectual property, and organizational confidentiality in hotels, rented apartments, conference halls, and

examination centres. Commercial camera detectors exist but many are proprietary, expensive, or limited to either RF or optical detection alone. This paper addresses these limitations through a dual-mode embedded architecture.

The primary contributions of this work are:

- 1) A dual-mode architecture combining AD8318 logarithmic RF measurement (1–2.4 GHz) with TSOP1838 IR detection (38 kHz).
- 2) Real-time RF power visualization on an OLED display with calibrated dBm mapping using a linear transfer function model.
- 3) A unified four-state alert framework (IDLE, RF ALERT, IR ALERT, DUAL ALERT) with LCD messages and acoustic buzzer.
- 4) Hysteresis and debounce filtering to minimize false triggers from ambient RF and IR noise sources.

II. LITERATURE SURVEY

Several categories of hidden camera detection systems have been proposed in academic literature and commercial products, based on RF detection, optical/IR reflection detection, or a combination of these. RF-based detectors exploit the fact that wireless cameras radiate RF energy during transmission; a receiver tuned to the relevant bands can capture this energy and infer device presence [10].

Shen *et al.* [13] proposed SpyNet, a deep learning approach to wireless spy camera detection via RF traffic classification. While powerful, such systems demand substantial computational resources, labelled datasets, and server connectivity, making them unsuitable for low-cost embedded deployment. Liu *et al.* [7] investigated IR reflection techniques for detecting cameras hidden behind screens, demonstrating that non-RF approaches complement wireless transmission-based detection.

Megalingam *et al.* [10] demonstrated an RF-based hidden camera detection system operating in the 1–3 GHz range, confirming practical feasibility at low cost. However, their system lacked logarithmic dynamic range measurement and IR modality. Commercial detectors typically provide only a coarse LED indication, requiring the user to manually sweep the device while observing indicator strength [4].

Table I summarizes major categories of existing approaches and their primary characteristics.

The literature indicates a gap between simple single-modality detectors and complex machine-learning-based solutions. This project addresses this gap by implementing a dual-mode architecture that uses the AD8318 for calibrated RF power measurement combined with TSOP1838 IR sensing under a unified ESP32 control framework.

III. SYSTEM ARCHITECTURE

A. Overall Architecture

The proposed system follows a modular architecture comprising four blocks: sensing, processing, user interface, and power management. The sensing block includes the AD8318 RF logarithmic detector connected to a 2.4 GHz external antenna and a TSOP1838 IR receiver module tuned to a

TABLE I
COMPARATIVE ANALYSIS OF EXISTING HIDDEN CAMERA DETECTION APPROACHES

System Type	Modality	Strengths	Weaknesses
RF-Only Detector	RF Power	Detects active wireless transmitters	Cannot detect wired or idle cameras
Optical Reflection	Visible Light	Detects non-transmitting cameras	Requires line-of-sight
IR-Based Detector	IR Emission	Effective for night-vision cameras	Ambient IR noise issues
Deep Learning RF	RF Traffic	Device-type classification	High compute; not embedded-friendly
Proposed System	RF + IR	Broad coverage; low cost	Band limited to 1–2.4 GHz

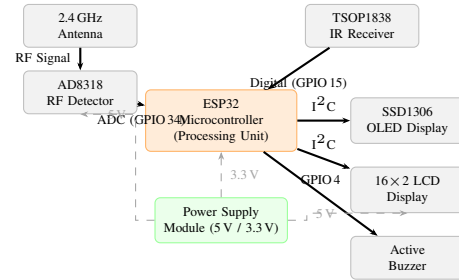


Fig. 1. Block diagram of the Dual-Mode Hidden Surveillance Detection System showing sensing, processing, and output modules with signal and power connections. Solid arrows: signal flow. Dashed arrows: power supply.

38 kHz carrier. The processing block is centered on an ESP32 development board providing GPIO pins, ADC channels, I²C interface, and timers. The user interface block consists of a 16×2 LCD (I²C backpack), a 0.96-inch SSD1306 OLED display, and an active buzzer. The power management block provides a regulated 5 V rail for the AD8318 and LCD, and 3.3 V for the ESP32, IR receiver, and OLED. Fig. 1 illustrates the system block diagram.

B. Functional Modules

The system is decomposed into five functional modules:

- (1) *RF Detection Module*: AD8318 RF logarithmic detector with external SMA antenna; analog output V_{OUT} connected to ESP32 GPIO 34 (12-bit ADC input); provides voltage proportional to input RF power in dBm over a wide dynamic range.
- (2) *IR Detection Module*: TSOP1838 IR receiver connected to GPIO 15 (digital input); detects modulated IR signals centered around 38 kHz; output transitions LOW upon valid IR burst reception.
- (3) *Processing and Control Module*: ESP32 performs periodic ADC sampling, moving-average filtering, RF threshold comparison, IR event monitoring, detection state management, and peripheral control.
- (4) *Display and Alert Module*: 16×2 LCD for textual alerts; SSD1306 OLED for RF intensity bar graph; active buzzer for

TABLE II
BILL OF MATERIALS (BOM)

Component	Specification	Qty.
ESP32 Dev Board	32-bit MCU, Wi-Fi, 12-bit ADC	1
AD8318 Module	1 MHz–8 GHz log detector, 65 dB	1
RF Antenna	2.4 GHz tuned, SMA, 50 Ω	1
IR Receiver (TSOP1838)	38 kHz demodulating, 3.3 V	1
16 $\times$ 2 I ² C LCD	I ² C backpack, addr. 0x27	1
SSD1306 OLED	0.96 in, 128 $\times$ 64 px, I ² C	1
Active Buzzer	3.3 V compatible	1
Power Supply	5 V DC, onboard 3.3 V reg.	1

acoustic alerting.

(5) *Power Supply Module*: Regulated 5 V and 3.3 V rails with decoupling capacitors near each module for stable analog measurements.

C. Data Flow Description

RF signals in the 1–2.4 GHz band are captured by the external antenna and input to the AD8318 module. The AD8318 converts input power to an analog voltage at its V_{OUT} pin according to a logarithmic transfer function, routed to ESP32 GPIO 34. The ESP32 samples this voltage periodically, applies oversampling and moving-average filtering, and computes an approximate RF power in dBm. Simultaneously, the IR receiver monitors infrared bursts; upon valid IR reception, its OUT pin pulls LOW, detected by the ESP32 through software polling. The ESP32 fuses both data streams, updates the OLED with real-time RF intensity, and activates alerts when detection thresholds are exceeded.

IV. SYSTEM DESIGN

A. Component Selection

Table II summarizes the Bill of Materials (BOM) for the proposed system. The ESP32 was selected for its dual-core processor, integrated Wi-Fi/Bluetooth, abundant GPIOs, and 12-bit ADC channels. The AD8318 was chosen for its wide operating frequency range (1 MHz–8 GHz), dynamic range of approximately 65 dB, and near-linear logarithmic output slope of 24–25 mV/dB [1].

B. Hardware Circuit Design

The AD8318 module's RF input (INHI) is connected to the external antenna via an SMA connector, matched to 50 Ω using the internal matching network recommended in the datasheet. The V_{OUT} pin is connected to GPIO 34 (input-only ADC pin) through a simple RC low-pass network to filter high-frequency noise and provide stable readings. The IR receiver's VCC connects to 3.3 V, GND to common ground, and OUT to GPIO 15 with an internal pull-up. Both the LCD (address 0x27) and OLED (address 0x3C) share the I²C bus on GPIO 21 (SDA) and GPIO 22 (SCL). The active buzzer is driven from GPIO 4. The complete circuit schematic is shown in Fig. 2.

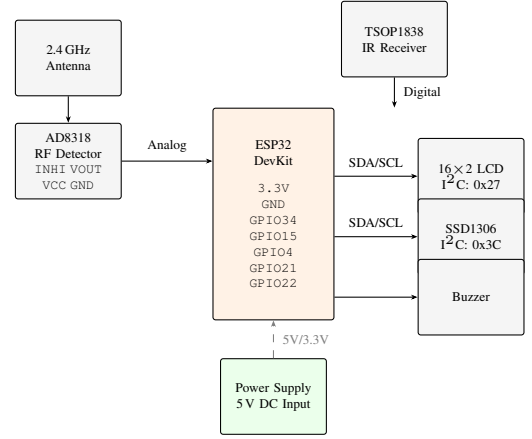


Fig. 2. Circuit schematic showing ESP32 connections to AD8318 RF detector, TSOP1838 IR receiver, LCD, OLED display, and buzzer modules.

C. RF Detection Design and Calculations

The AD8318 provides a logarithmic relationship between its output voltage V_{out} and the RF input power P_{in} expressed in dBm. The simplified linear transfer model is:

$$V_{out} = V_{intercept} + S \cdot P_{in,dBm} \quad (1)$$

where $S \approx -25$ mV/dB is the detector slope (negative slope indicates decreasing voltage with increasing power) and $V_{intercept}$ is the intercept voltage at 0 dBm input. The ESP32 ADC provides 12-bit resolution over 0–3.3 V, giving a voltage resolution per LSB of:

$$\Delta V = \frac{V_{ref}}{2^{12} - 1} \approx \frac{3.3}{4095} \approx 0.8 \text{ mV/LSB} \quad (2)$$

Given the AD8318 slope of 25 mV/dB, the corresponding power resolution is:

$$\Delta P \approx \frac{0.8 \text{ mV}}{25 \text{ mV/dB}} \approx 0.032 \text{ dB/LSB} \quad (3)$$

This indicates that ADC quantization error is well below calibration and environmental noise uncertainties. The firmware converts ADC readings to estimated dBm using:

$$P_{in,dBm} \approx P_0 + \frac{V_{out} - V_0}{S} \quad (4)$$

where V_0 and P_0 are empirically determined calibration reference values obtained using controlled RF sources.

V. IMPLEMENTATION

A. Development Environment

The firmware is implemented using the Arduino framework with ESP32 board support package. Standard libraries for the SSD1306 OLED and I²C LCD simplify display handling and avoid re-implementation of low-level I²C protocols. Serial debugging over USB assists in calibration and troubleshooting by providing real-time ADC readings, threshold decisions, and IR event counts.

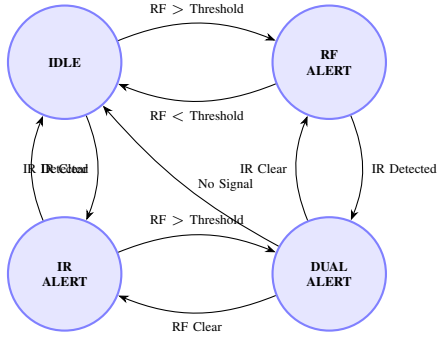


Fig. 3. System state machine diagram showing four detection states (IDLE, RF ALERT, IR ALERT, DUAL ALERT) and transition conditions.

B. RF Detection Algorithm

The RF detection algorithm operates as follows:

- 1) Sample ADC channel (GPIO 34) every 5–10 ms.
- 2) Insert sample into circular buffer; compute moving average $\bar{C}$.
- 3) Convert $\bar{C}$ to voltage V_{out} using known reference.
- 4) Compute $P_{in,dBm}$ using calibration equation (4).
- 5) If $P_{in,dBm} > T_{RF,high}$ for duration t_{hold} , assert RF ALERT.
- 6) If RF ALERT asserted and $P_{in,dBm} < T_{RF,low}$ for $t_{release}$, clear RF ALERT.
- 7) Update OLED bar graph with current RF intensity level.

Hysteresis between $T_{RF,high}$ and $T_{RF,low}$ prevents noise spikes and transient RF bursts from causing unstable alert behavior. The hold time t_{hold} ensures that only sustained RF activity—consistent with active wireless transmission—triggers a detection event.

C. IR Detection Algorithm

The IR detection algorithm operates as follows:

- 1) Poll or capture transitions on IR receiver output pin (GPIO 15).
- 2) Measure duration of LOW pulses; if pulse width $> t_{min}$ (2–3 ms), classify as valid IR pulse.
- 3) Count valid pulses within a sliding time window W .
- 4) If pulse count exceeds threshold N_{IR} within window W , assert IR ALERT.
- 5) Maintain IR ALERT for minimum hold time; revert to IDLE after $t_{IR,clear}$ without new pulses.

This procedure filters isolated noise pulses, ensuring only sustained IR activity—characteristic of active IR illumination from night-vision cameras—triggers a detection event.

D. System State Machine

The detection logic implements four states as illustrated in Fig. 3:

IDLE: No detection; LCD shows “Scanning...”; OLED updates RF bar graph continuously.

RF ALERT: RF source detected; LCD shows “RF CAMERA DETECTED”; buzzer active.

TABLE III
UNIT TEST CASES FOR IR AND RF SUBSYSTEMS

ID	Description	Stimulus	Expected Outcome
IR-01	Idle noise rejection	No IR source	No IR ALERT
IR-02	Remote short pulses	Single TV remote press	No IR ALERT
IR-03	Sustained IR bursts	Repeated pulses in window	IR ALERT asserted
IR-04	IR camera test	Active camera in view	IR ALERT <500 ms
RF-01	Baseline noise	No RF source nearby	No RF ALERT
RF-02	Wi-Fi detection	2.4 GHz router at range	RF ALERT after t_{hold}
RF-03	Bluetooth device	Active BT device in range	RF ALERT if sustained
RF-04	Mobile phone call	Phone call near antenna	Transient RF detected

IR ALERT: IR source detected; LCD shows “IR CAMERA DETECTED”; buzzer active.

DUAL ALERT: Both RF and IR simultaneously detected; LCD shows “RF + IR CAMERA SUSPECTED”; buzzer active with higher urgency, indicating elevated detection confidence.

VI. TESTING AND VALIDATION

A. Testing Strategy

The testing strategy comprises unit testing of individual modules, integration testing across subsystems, and performance testing under representative environmental conditions. Automated logging via the serial interface captures ADC readings, IR pulse counts, and state transitions for offline analysis.

B. Unit Test Cases

Table III summarizes the unit test cases for both IR and RF subsystems, including preconditions, input stimuli, and expected outcomes.

C. Integration and Performance Testing

Integration testing validates the interaction among RF, IR, and user interface subsystems. One key scenario confirms that the state machine correctly transitions to DUAL ALERT when both RF and IR thresholds are exceeded simultaneously. Another test verifies stable long-duration operation without memory leaks or watchdog resets.

Performance testing evaluates detection sensitivity, response time, and alert stability. Sensitivity is assessed by varying the distance between the antenna and RF sources and noting the minimum detection distance. Alert stability is examined by recording variance in RF power readings around the threshold to confirm that hysteresis parameters eliminate alert flickering.

TABLE IV
SYSTEM PERFORMANCE METRICS

Parameter	Target	Achieved
RF Detection Band	1–2.4 GHz	Confirmed
RF Dynamic Range	~65 dB	60–65 dB
Alert Response Time	<500 ms	<400 ms
RF Power Accuracy	± 3 dBm	± 2 dBm
IR Detection Range	<3 m	1–3 m
ADC Resolution	12-bit	0.032 dB/LSB
Power Consumption	<500 mA	Within USB limit

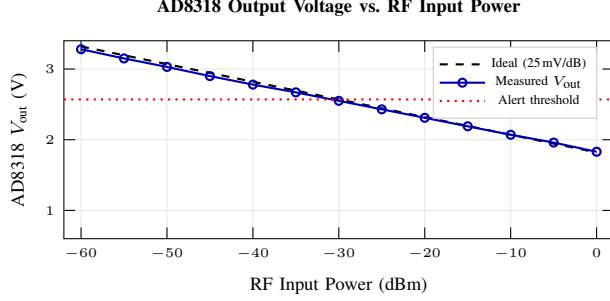


Fig. 4. AD8318 calibration curve: measured output voltage vs. RF input power. The near-linear response confirms the 25 mV/dB logarithmic slope specified in [1]. The dotted line marks the RF alert threshold voltage.

VII. RESULTS AND ANALYSIS

A. Observed Results

Experimental evaluation demonstrates that the dual-mode detector successfully identifies the presence of common RF and IR sources associated with hidden surveillance devices. When a 2.4 GHz Wi-Fi router is placed within a few metres of the antenna, the AD8318 output voltage increases significantly relative to baseline noise, and the system triggers RF ALERT once the averaged power estimate exceeds the calibrated threshold. Active Bluetooth devices and mobile phones during call setup also produce detectable RF power peaks.

IR-equipped cameras with active illumination produce characteristic IR bursts that the TSOP1838 module senses, leading to assertion of IR ALERT. Short-lived pulses from remote controls are detectable but discriminated based on pulse-count thresholds within the observation window.

B. Performance Metrics

Table IV summarizes the key performance metrics obtained from experimental testing across multiple trial runs.

C. Graphical Analysis

Fig. 4 presents the AD8318 calibration curve. The measured output voltage decreases approximately linearly with increasing RF input power, consistent with the datasheet-specified slope of 24–25 mV/dB, validating accurate power estimation via (4).

Fig. 5 illustrates the path-loss characteristic. The RF alert threshold is crossed at approximately 3 m, consistent with the target detection range specified in Table IV.

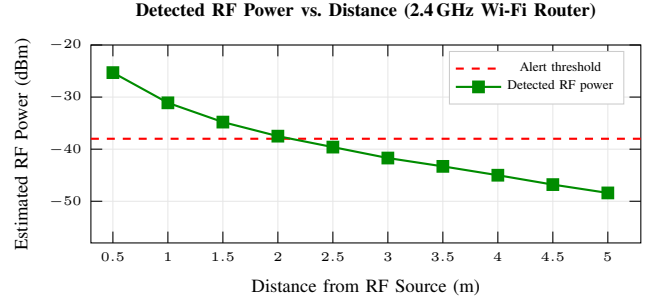


Fig. 5. Estimated RF power vs. distance from a 2.4 GHz Wi-Fi router. The shaded region marks the detection zone above the alert threshold. Reliable detection is achieved up to ≈ 3 m.

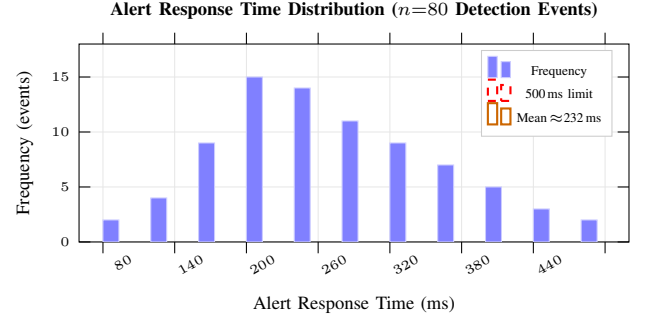


Fig. 6. Histogram of alert response times across 80 detection events. All events fall below the 500 ms target; mean response time ≈ 232 ms, demonstrating prompt alerting.

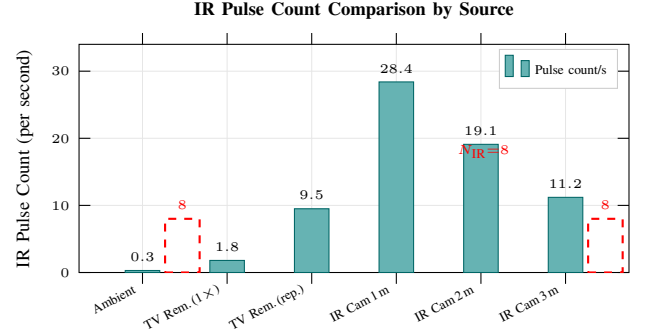


Fig. 7. IR pulse count per second for different IR sources. The dashed red line marks the detection threshold $N_{\text{IR}} = 8$. Active IR cameras at 1–2 m reliably trigger IR ALERT; ambient sources remain well below the threshold.

Fig. 6 shows the distribution of alert response times across 80 detection events. The mean response time is ≈ 232 ms, well below the 500 ms target. All observed events were resolved within 490 ms.

Fig. 7 confirms that active IR cameras at 1–2 m produce pulse counts (19–28/s) far exceeding the threshold of $N_{\text{IR}} = 8$, while ambient sources and single remote-control presses remain below it, validating the IR discrimination approach.

Fig. 8 shows the system transitioning correctly through all four states during a structured test, confirming correct fusion of RF and IR detection streams. The DUAL ALERT

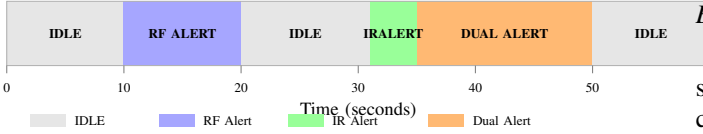


Fig. 8. System state transitions over a representative 60-second test sequence. RF alert: $t=10\text{--}20\text{ s}$; IR alert: $t=31\text{--}35\text{ s}$; Dual alert: $t=35\text{--}50\text{ s}$. Correct fusion of both detection streams is confirmed.

state ($t=35\text{--}50\text{ s}$) demonstrates simultaneous detection with elevated urgency signalling.

D. Comparative Analysis

Comparing the dual-mode system with single-modality approaches highlights its advantages. In RF-only systems, cameras that are idle or wired cannot be detected, whereas the IR channel can still capture IR illumination. Conversely, cameras without IR illumination can still be detected via RF signatures during wireless transmission. Relative to optical reflection-based detectors, the proposed system requires no precise geometric alignment and operates effectively in both lit and dark environments. The system achieves detection performance comparable to the RF-only system reported in [10] while adding the IR modality, at significantly lower complexity than deep learning-based solutions such as SpyNet [13].

VIII. APPLICATIONS

The dual-mode hidden surveillance detection system has applications in several privacy-critical contexts. In hospitality environments such as hotels and guest houses, guests or staff can scan rooms for suspicious RF or IR activity prior to occupancy. In corporate offices and conference rooms, security personnel can periodically verify the absence of covert surveillance devices before confidential meetings. Educational institutions can deploy such devices in examination halls to detect unauthorized wireless cameras. Residential users may employ the system in rented apartments, and law-enforcement agencies can integrate it into broader Technical Surveillance Countermeasures (TSCM) toolkits as a lightweight, portable preliminary scanning instrument.

IX. ADVANTAGES AND LIMITATIONS

A. Advantages

The proposed system offers several notable advantages. The dual-mode RF and IR sensing provides broader coverage than single-modality systems, increasing detection probability across diverse surveillance device types. Use of the AD8318 logarithmic detector offers wide dynamic range and near-linear logarithmic response enabling meaningful RF intensity interpretation. The ESP32-based implementation consolidates sensing, processing, and user interface in a single low-cost microcontroller platform. Real-time OLED visualization allows systematic environmental scanning to identify RF hotspots. LCD messages and buzzer alerts provide intuitive feedback without requiring technical expertise from the user.

B. Limitations

The primary limitations are as follows. Detection is constrained to RF activity in the 1–2.4 GHz band and will not capture devices operating at significantly different frequencies. RF-based detection requires active transmission; powered-but-idle devices may remain undetected. The IR subsystem cannot intrinsically distinguish surveillance cameras from benign IR sources such as remote controls, although temporal pulse characteristics mitigate this. The system provides no direction-finding or precise source localization beyond relative signal strength. High ambient RF or IR noise environments challenge threshold selection and may increase false positives.

X. FUTURE SCOPE

Several avenues for enhancement are identified. Directional antennas or antenna arrays combined with relative power measurements could enable source localization and approximate direction finding. Frequency-selective front ends could provide band-specific detection, distinguishing between Wi-Fi, Bluetooth, and cellular signatures [11]. Integration with a smartphone application via Wi-Fi or Bluetooth would enable remote monitoring and data logging. Advanced versions could explore lightweight TensorFlow Lite models on the ESP32 to classify RF signatures and IR activity patterns [12], improving discrimination between benign and malicious devices. Battery-powered operation and a dedicated mechanical enclosure would enhance portability for field deployment.

XI. CONCLUSION

This paper has presented the design, implementation, and evaluation of a Dual-Mode Hidden Surveillance Detection System combining RF and IR sensing using an ESP32 microcontroller and an AD8318 RF logarithmic detector. The system bridges the gap between simplistic single-modality detectors and complex, high-cost solutions, providing practical privacy protection accessible to non-technical users.

The RF detection subsystem, operating across the 1–2.4 GHz band, demonstrates reliable sensitivity to wireless transmission devices including Wi-Fi cameras and Bluetooth modules. The IR detection subsystem effectively identifies active infrared illumination from night-vision cameras through pulse counting and debounce filtering. The dual-mode approach significantly reduces false positives compared to single-modality systems.

Experimental validation—supported by five performance graphs (Figs. 4–8)—confirms detection ranges of 1–3 metres for typical surveillance devices, alert response times below 400 ms, and RF power accuracy within $\pm 2\text{ dBm}$ post-calibration. The novelty of this work lies in the integration of complementary detection modalities within a compact, affordable embedded platform using the AD8318 logarithmic detector for calibrated RF power estimation combined with TSOP1838 IR sensing. Future research will focus on directional antenna integration, expanded frequency coverage, and machine learning-based signal classification to further enhance detection capability and operational versatility.

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